Assignment 11 — Team Affordance Report

Team Member	Ontology Modeling	Additional Responsibilities
Alex		
Hyejin	Conceptual model design and diagrams; feasibility semantics, SWRL rule revisions, and core ontology maintenance	Ontology metadata/versioning, provenance and citation review, Curated Term List–RDF alignment, integration/QA, and report consolidation
Xiye	CQ1 Improvement.	Term List Update, Website Update
Lia	CQ2 query revisions (Q2.1-Q2.4), merged CQ4 query (Q4.1), new CQ5 query (Q5.1); validation of CQ2 individuals with the OE_11 namespace and renamed SWRL rules	Use Case updates (CQ2, CQ4, CQ5, rule names, open issues) addressing Daniela's feedback; final merged OE_11 query file (ind: namespace, all queries rerun); Report sections 1.2 CQ2, 2.1, 3 and 4

1. Ontology and Query Updates

1.1 Ontology Updates — Hyejin

Feasibility Reasoning and SWRL Updates

Based on Assignment 10 feedback, we refined the feasibility reasoning model by introducing `hasUnresolvedRequirement` to distinguish unresolved requirements from satisfied and conflicting ones. We also revised the SWRL rules to handle unspecified rigidity, edge thickness, or both, making insufficient-information judgments more explicit.

Ontology Metadata and Annotations

We updated the metadata, provenance, and annotations of both the Main and Individuals ontologies, including definitions, notes, examples, and version information. We synchronized the Curated Term List with the RDF files, standardized individual namespaces and SWRL rule labels, and revised the Conceptual Model documentation and diagrams to reflect the OE11 changes while preserving the existing class structure.

1.2 Competency Questions and Query Updates

CQ1 — Xiye

- Query revisions and additional queries: Shortened Q2 for CQ1 with Danielle's feedback considered. Deleted redundant information.Kept only the necessary data chart.
- Changes to individuals and expected results: No new individuals or expected results need to be added for my part. One minor change might be relevant: the expected result of Q2 for CQ1 hasn't changed, but the query was shortened to match the expected result.

CQ2 — Lia

- **Query revisions and additional queries:** Following Daniela's Assignment 10 feedback, Lia updated all CQ2 queries for the OE11 `ind:` namespace and documented each query's English question, required Protégé tool, and expected behavior without a reasoner. Q2.1 retrieves inferred judgments for the card, rolling pin, sponge, and spatula after HermiT. Q2.2 was rewritten to retrieve satisfied, conflicting, and unresolved requirements using inferred assessment-level properties. Q2.3 compares required and actual feature levels without CONCAT, BIND, or GROUP BY, while Q2.4 lists judgments for all carrot assessments. She also merged two CQ4 queries into Q4.1 (card vs. sponge) and added Q5.1 to identify undecidable cases, satisfied requirements, and missing features.
- **Changes to individuals and expected results:** No new CQ2 individuals were added. The existing spatula candidate was integrated under the `ind:` namespace. HermiT results confirmed that the card is feasible, the rolling pin and sponge are infeasible, and the spatula has insufficient information, with an inferred `hasUnresolvedRequirement` for edge thickness despite satisfying rigidity. The merged ontology remained consistent, with no unsatisfiable classes.

2. Challenges and Remaining Issues

2.1 SPARQL Query Challenges — Xiye, Lia, Alex

- Problems encountered during query development or testing

  Lia (CQ2): Daniela's feedback said Q2.2 did not return the expected answer in Assignment 10: it mixed inferred assessment results with the asserted hasUnspecifiedFeatureType marker on the object. It was rewritten to read only the inferred assessment-level properties (satisfiesRequirement, conflictsWithRequirement and hasUnresolvedRequirement) through a UNION. Q2.3 was simplified to avoid CONCAT, BIND and GROUP BY. The standard SPARQL Query tab only sees asserted triples, so every query now states whether it needs Snap SPARQL after HermiT or the standard tab, and what it returns without a reasoner. Moving the individuals to the ind: namespace also required updating every individual IRI in the queries. Because the Assignment 10 file was submitted before the final runs were checked, every query in the final OE_11 team file was run end to end on the OE_11 files, both on the asserted triples and on the HermiT entailments, and its EXPECTED block now records that result.

  Xiye (CQ1): Not many problems.

- Remaining limitations or unresolved issues

Lia (CQ2): Insufficient Information still relies on an explicitly asserted hasUnspecifiedFeatureType marker; a feature value that is simply absent is not detected under open-world semantics. The CQ2 entailment queries also cannot be checked in the standard SPARQL Query tab, so their results depend on running HermiT before Snap SPARQL.

Xiye (CQ1): …

2.2 Ontology Annotation and Integration — Hyejin

The assignment reinforced that explicit conflict and missing information are different under the open-world assumption. It also showed that conceptual diagrams, formal axioms, annotations, and test data must be maintained as one coherent specification: documentation can become misleading even when the RDF remains syntactically valid. Provenance is not just filling a source field; the source must support the meaning actually represented. In addition, diagrams that separate asserted and inferred class relationships helped us explain what OWL alone establishes versus what SWRL rules infer during assessment.

3. Lessons Learned

- **Hyejin:** This assignment reinforced the importance of distinguishing explicit conflicts from missing information under the open-world assumption. It also highlighted the need to keep conceptual models, OWL/SWRL reasoning, annotations, and test data consistent, while ensuring provenance accurately supports the represented concepts.
- **Lia:** I learned that query results depend on the reasoning environment: Snap SPARQL after HermiT can retrieve inferred judgments that standard SPARQL cannot. Clearly documenting query requirements and explicitly modeling missing information improved testing and interpretability.
- **Xiye:** I learned the importance of presenting information concisely to reduce the audience's cognitive load. Maintaining provenance across the Term List and ontology also highlighted the challenges of keeping ontology documentation consistent.

4. AI Usage

- **Hyejin:** ChatGPT assisted with annotation wording, citation checking, documentation, and RDF synchronization. Suggestions were reviewed against ontology requirements, and RDF parsing and graph comparisons verified structural preservation.
- **Lia:** Claude assisted with drafting and testing CQ2, CQ4, and CQ5 queries using rdflib and HermiT, as well as Use Case suggestions and report writing. All outputs were reviewed, and Use Case changes were submitted as suggestions.
- **Xiye:** ChatGPT assisted with Usage Notes for High, Medium, and Low. All other work was completed manually. link: https://chatgpt.com/share/6af775ed-c518-83ea-84b4-13ff25c6433e